

Digital Video Stabilization Method Based on Periodic Jitters of Airborne Vision of Large Flapping Wing Robots

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Abstract—Large-scale flapping wing robots (FWRs) with airborne vision have important applications in visual navigation, aerial surveying, fire warning and power-line inspection. However, airborne vision and its videos suffer from strong jitters due to periodic wing flapping, which lowers the success rate of detection and measurement precision. In this paper, a robust digital video stabilization (DVS) method based on periodic jitters is proposed to provide continuous stable monitoring video without pan-tilt camera assistance. First, the periodic motion model of the FWR is established for video jitter analysis. Second, jitter frequencies in different flight states are estimated by continuous jitter acceleration. Then, feature trajectories generated from the video are adjusted adaptively for jitter frequency consistency and smoothed individually by the sampling-interpolation-averaging strategy, including the short trajectories. The stabilized video is generated by guidance from the original and smoothed trajectories. Finally, the proposed method is tested in outdoor flights with a 2.2-meter wingspan FWR and is found to outperform traditional, commercial, and deep learning DVS methods in terms of stability and robustness in various scenes and flight states.

Index Terms—Airborne vision, flapping wing robot, video monitoring, video stabilization.

I. INTRODUCTION

LARGE flapping wing robots (FWRs) have received much attention in [1] and [2] as a novel type of bionic flying robot with superior load capacity and durability compared to small-scale robots. They also have more advantages than unmanned aerial vehicles (UAVs) in camouflage because of their bionic shape and flight style. With the help of airborne

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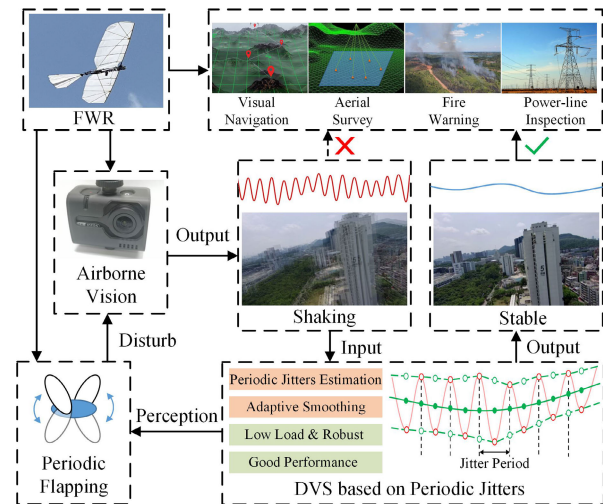


Fig. 1. The motivation and effect of the proposed method for FWR applications.

vision, there are important applications for large-scale FWR in measurement and detection as well as for other UAVs, such as visual navigation, aerial surveying, fire warning, and power-line inspection. The detection success rate and measurement accuracy in these tasks depend on a stable video stream, which FWRs find difficult to maintain due to the strong jitters caused by periodic wing flapping, as shown in Fig. 1.

In contrast to small-scale FWRs, which could hover in [3], large-scale FWRs have difficulty maintaining body balance due to low-frequency flapping resulting in strong jitters. The authors of [4] noted that the periodic average theory of small-scale FWRs may not be suitable for large-scale FWRs. The Griffin group in [5] and [6] applied an event camera to their large-scale FWR for autonomous obstacle avoidance and found that the jitters of the FWR were much stronger than those of the quadrotor. Video captured by airborne vision was shaky during the flight of FWR, which induced unpredictable risks to measurement and detection. Therefore, it is necessary to use video stabilization to address the difficulties introduced by high-speed flight and quick turning of large-scale FWRs.

A popular method for video stabilization is to use a gimbal or pan-tilt camera in [7], [8], and [9]. However, these would increase the mass, volume, and power consumption of the FWR, and their structural reliability may also be weak. In addition, due to their limited rotational range and low degrees

of freedom, it is difficult for gimbal cameras to totally eliminate strong and complicated jitters, and their view would be unstable during FWR turning quickly.

Digital video stabilization (DVS) is another method that does not require additional hardware assistance and is implemented only by image algorithms. DVS based on two-dimensional (2D) transformation is robust and efficient in [10] and [11] and could achieve real-time processing in [12] and [13] but induces significant artifacts and distortion in videos with strong parallax. Three-dimensional (3D) reconstruction and image rendering have been adopted for DVS to handle strong parallax in [14], [15], and [16], while its computing efficiency and reliability are also weakened.

DVS based on feature trajectories could stabilize videos with strong parallax efficiently in [17], [18], [19], and [20] by smoothing the feature trajectories from videos and warping frames by content-preserving warps (CPW) in [21]. The authors of [18] noted that Gaussian smoothing is not suitable for discontinuous trajectories and smoothed them in a low-dimensional linear space. The authors of [19] smoothed trajectories by minimizing one third of their derivatives. In [20], the trajectories are smoothed by time and spatial constraints from the Bessel curve and Delaunay triangulation. However, these methods may fail when there are many short trajectories, which appear during FWR turning.

The authors of [22] smoothed virtual trajectories generated by epipolar geometry instead of real trajectories, but the virtual trajectories could not handle the case of camera rotation. The authors of [23] used low-rank matrices to complete short trajectories, but this method is time-consuming and does not have strong results. The authors of [24] stabilized video by using a bundle of multiple camera paths and space-time path optimization, which made it difficult to maintain the balance of stability and cropping rate of video. In the case of foreground occlusion, a hybrid strategy was adopted in [25], which used 3D stabilization for foreground objects and 2D stabilization for the background. The authors of [26] and [27] used the joint optimization of feature trajectories from the foreground and background. However, these methods are not efficient, and there is little foreground occlusion in aerial videos. The authors of [28] and [29] used independent component analysis and the Hilbert–Huang transform to remove jitters in video. However, this method relied on long-time local motions and could not be used to decompose complex coupled motions.

In recent years, an increasing number of studies have applied deep learning to DVS, and there are two main research directions. The first is based on extensive training with labeled data; one method that uses this strategy is StabNet [30], [31], which achieved real-time stabilization. These methods are mostly used for handheld video, but it is impossible for FWRs to obtain enough labeled data. The second is unsupervised, non-long-term training, which was used in [32], DIFRINT [33] and DUT [34]. These methods have better performance than the traditional methods, but they are still time-consuming even with GPUs, which makes it difficult to meet the process time and hardware requirements of FWR.

TABLE I
COMPARISON OF VIDEO STABILIZATION METHODS

Method	Gimbal Camera	Subspace[18]	DUT[34]	Proposed
Characteristic	Mechanical	Linear Space	Deep Learning	Periodic Jitters
Load	P	G	G	G
Stable View	P	G	G	G
Efficiency	G	M	P	M
Robust	M	P	G	G
Stability	M	M	G	G

G:Good M:Medium P:Poor

For visual measurement and detection by large-scale FWRs, a DVS method based on periodic jitter is proposed to provide continuous and stable monitoring video without pan-tilt cameras or other hardware. There are three main contributions:

- The jitter frequency is estimated by vision based on the characteristics of periodic wing flapping, and feature trajectories from video are adjusted adaptively for jitter frequency consistency. Videos in various flight states can be optimally stabilized without manipulating the parameters.
- The feature trajectories are smoothed by the strategy of sampling-interpolation-averaging, and the sampling interval is also automatically regulated to fit short trajectories. All the feature trajectories can be smoothed individually, including the short trajectories, which makes this method efficient and robust.
- The whole process of video stabilization is implemented and tested in outdoor flights with a 2.2 m wingspan FWR in straight-flying, turning, and unsteady states. The method performs well according to the subjective and objective evaluations. The links of related videos are provided in the appendix.

As a summary, Table I shows a comparison of different video methods, which are further illustrated and validated in the following. Section II describes the FWR monitoring system and stabilization process. Section III describes the jitter frequency estimation method. Section IV describes the adjustment and smoothing of feature trajectories. The stabilization results are evaluated and compared in Section V. Finally, the conclusions and future work are presented.

II. FWR MONITORING SYSTEM & VIDEO STABILIZATION PROCESS

The FWR monitoring system includes a large-scale FWR and airborne vision, which has good wind resistance and endurance.

A. Parameters of FWR

The wingspan of the adopted FWR is 2.2 m, the empty endurance is 90 min and the maximum load is 200 g, while the maximum load installation space is 100 mm×100 mm×100 mm, as shown in Fig. 2. There are

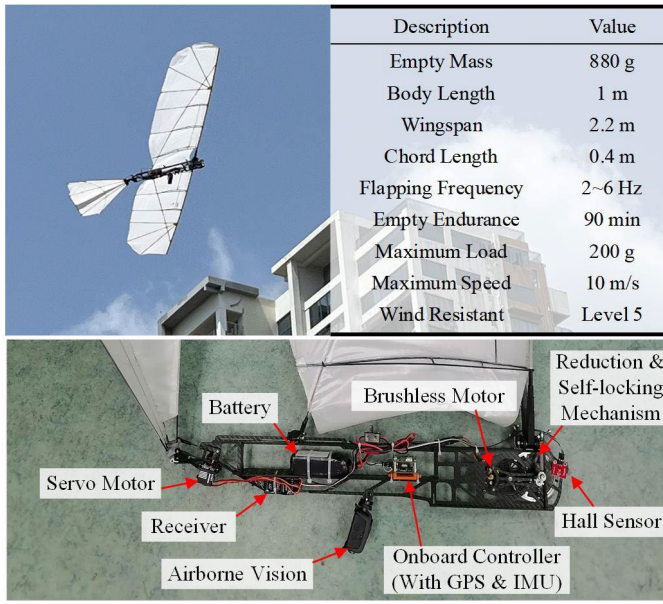


Fig. 2. The adopted FWR and airborne vision.

three parts of the FWR: body, wings and tail. The lightweight, airfoil-shaped wings can produce enough lift and thrust force. The tail has two degrees of freedom: pitch and roll, which can be used to adjust the attitude and direction of flight.

Wing flapping is the primary flight style of FWR, and it can be divided into straight or turning flight according to flight path. The tail can enhance the motility of the FWR, such as when turning quickly with a small radius for obstacle avoidance. The FWR also has a remote controllable self-locking mechanism, which can lock the wings for gliding and improve the endurance. When the FWR is climbing, the wing can be unlocked for flapping.

B. Control of FWR

The control of FWR includes an autonomous mode with onboard controller and manual mode with remote control. In the autonomous mode, the onboard controller receives task commands from the ground workstation and outputs control signals for the expected altitude, pose and path, according to the position and attitude data obtained by the Global Positioning System (GPS) and inertial measurement unit (IMU). The brushless motor and servo motors receive these signals and drive the tail and wings of the FWR for flying straight, turning, gliding and climbing in the specified pose and flapping frequency. The video is obtained by airborne vision and then sent back to the ground workstation, where it is stabilized offline for visual tasks such as measurement, detection or tracking. The positions of these components are shown in Fig. 2, and the process above is also shown in Fig. 3.

C. Airborne Vision

The airborne vision is a small-scale camera (XTU S2 Camera), which is regarded as an additional load. The mass is approximately 98 g, with a resolution of 1280×720 and frame rate of 60. The camera is mounted and fixed on the

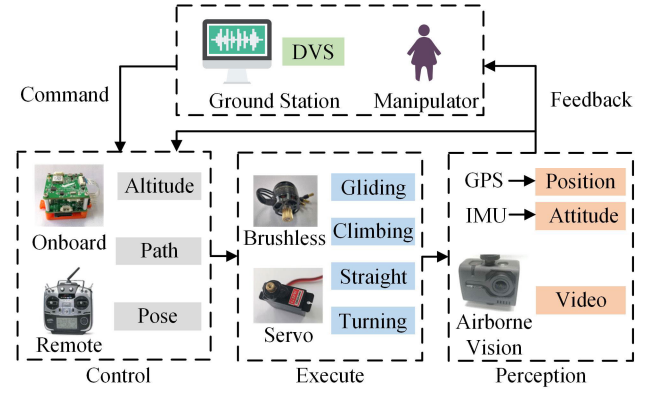


Fig. 3. The control process of FWR.

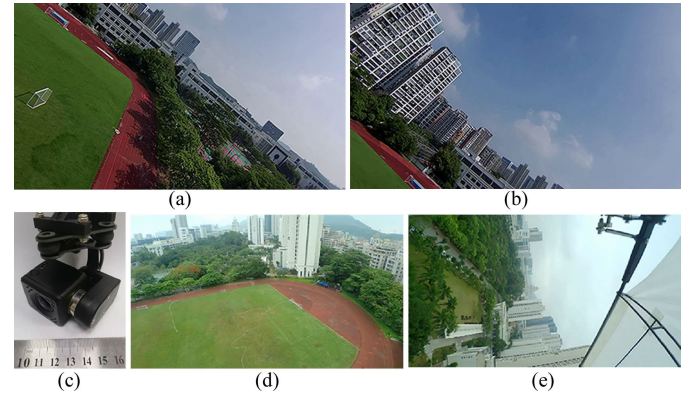


Fig. 4. The scenes of video before (a) and after (b) the FWR quick turning in 0.5 s. (c) Small scale gimbal camera. The normal view (d) of gimbal camera and the unsteady view (e) while turning.

body of the FWR, which is powered by its own battery, and the maximum shooting time is approximately 2 hours.

During outdoor flight, the monitoring video of static buildings and vegetation is captured by airborne vision. The videos mentioned in the following are of this kind. The airborne vision is disturbed by wing flapping, resulting in strong video shaking. On the other hand, the scenery of the video may change quickly because of quick turning, as shown in Fig. 4(a)(b), which may induce many short feature trajectories.

For video stabilization, a small-scale gimbal camera (Ark-bird Mini Gimbal Camera) was tested, as shown in Fig. 4(c), but the performance was not satisfactory. The gimbal camera could not eliminate video jitters completely, and the stability was poor in cases of intense motion. As shown in Fig. 4(d)(e), the view of the gimbal camera may suddenly change while the FWR turns quickly. For a stable view, a camera that could be mounted and fixed was adopted instead of a gimbal camera.

The reader is encouraged to watch the videos captured by the gimbal camera; their links are shown in the appendix.

D. Video Stabilization Process

For stable monitoring videos, the process of the proposed method is established, as shown in Fig. 5. First, the feature trajectories are generated from the original video by the Kanade–Lucas–Tomasi (KLT) tracker in [35]. The jitter frequency is estimated based on global jitter acceleration

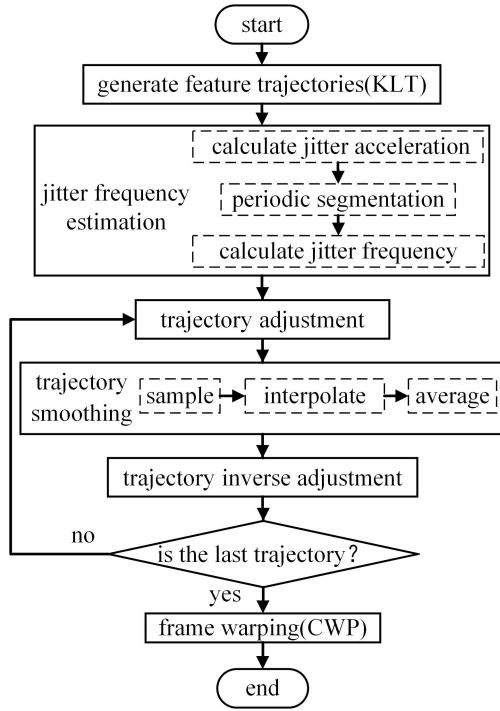


Fig. 5. The flowchart of proposed method.

by peak detection and periodic segmentation. Second, the time intervals of feature trajectories are adjusted temporarily for frequency consistency. Then, the sampling-interpolation-averaging strategy is used to smooth feature trajectories individually, including short trajectories. The smoothing results are adjusted inversely. Finally, with the guidance of original and smooth trajectories, the stabilized video is generated by content-preserving warps (CPWs) as in [21].

III. FREQUENCY ESTIMATION & TRAJECTORY ADJUSTMENT

Video jitters of airborne vision are closely related to FWR periodic wing flapping, which is different from the random jitters of hand-held cameras. Thus, the jitter frequency can be estimated visually.

A. Jitter Analysis

The shifting acceleration of the FWR body and airborne vision causes video jitters. The largest contributor to this force imbalance is wing flapping, which is the only source of power. Assuming that the FWR is flying horizontally in its ideal state, the force analysis of the FWR body during periodic wing flapping is shown in Fig. 6(a).

The body is affected by the thrust F_T and air resistance F_R horizontally and the flapping force F_L and gravity G vertically. According to the Knoller–Betz effect in [36], the wing flapping cycle is divided into downstroke and upstroke. The wings flap downward during the downstroke, producing an upward F_L . During the upstroke, the wings flap upward, while the F_L is downward. Hence, during a single cycle, the directions of F_T and F_R remain steady, but the directions of F_L change up and

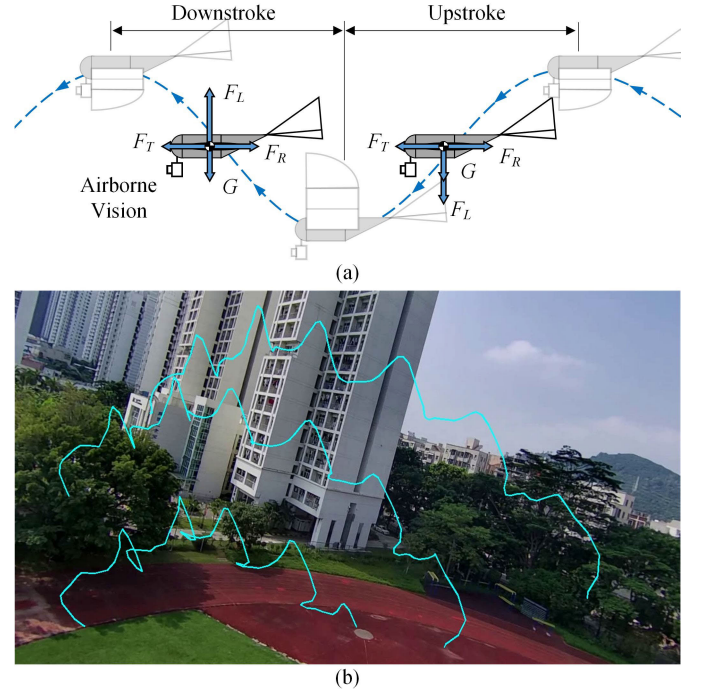


Fig. 6. (a) Affected by wing flapping, the FWR body would shake periodically. (b) The feature trajectories of video form airborne vision with periodic jitters.

down periodically. The acceleration of the FWR body in the vertical direction is denoted as a_J :

$$a_J \propto F_L - G \quad (1)$$

a_J also changes periodically following F_L , inducing periodic and strong jitters in airborne vision and its video. When F_L is in the same direction as G in the downstroke, a_J reaches its maximum. Therefore, the jitter frequency and amplitude change according to periodic flapping. When the FWR stops flapping and glides, the jitters disappear. In contrast, they become much stronger while climbing.

In addition, there are other high-frequency but minor jitters in video caused by mechanical and environmental factors. In the following, it is found that these jitters are also smoothed while smoothing the main ones. For simplification, periodic wing flapping is considered the main cause of video jitters, which is the basis of jitter frequency estimation.

B. Jitter Frequency Estimation

The jitter frequency of video can be estimated visually. First, feature trajectories of video are obtained by the KLT tracker, which can quantitatively describe the actual motion of airborne vision. Several real feature trajectories are shown in Fig. 6(b), which show some distinct periodic shapes, such as peaks and valleys. Although the length, amplitude and direction of these feature trajectories may differ from each other due to the changing scenery and the movement of the FWR, they all have the same jitter frequency at the same time.

The global jitter acceleration of video is a , while $a \propto a_J$, which is undisturbed by the discontinuous trajectories. Assuming that the t th frame of video has n feature points,

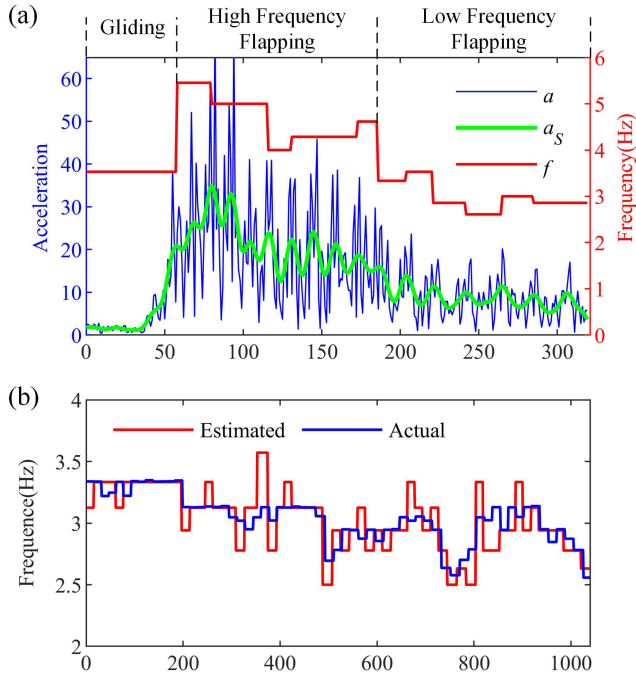


Fig. 7. (a) The jitter acceleration and frequency of video in flight states from gliding to flapping. (b) The comparison between the estimated jitter frequency and actual flapping frequency.

the pixel coordinates of the i th ($i \in [1, n]$) point are (x_{it}, y_{it}) , while its pixel coordinates in the $t-1$ th and $t+1$ th frames are (x_{it-1}, y_{it-1}) and (x_{it+1}, y_{it+1}) . Since the jitter acceleration of each feature point is different, a is calculated by:

$$a(t) = \frac{1}{n} \sum_{i=1}^n \sqrt{(x_i^{t+1} + x_i^{t-1} - 2x_i^t)^2 + (y_i^{t+1} + y_i^{t-1} - 2y_i^t)^2} \quad (2)$$

To remove the high-frequency noise in a , mean smoothing and Fourier smoothing are adopted, and the smoothed results a_s are generated. According to the minimum flapping period of the FWR, the mean smoothing radius is set to 20%~25% of the minimum flapping period, and the Fourier filter threshold is set to 120%~140% of the maximum flapping frequency to avoid oversmoothing. For example, the minimum flapping period is 10 in this work, as the maximum flapping frequency is 6 Hz and the video frame rate is 60. Therefore, the radius is set to 2, and the threshold is set to 8 Hz.

The a of an original video sample in flight states from gliding to flapping is shown in Fig. 7(a). Although the frequency and amplitude are not strictly the same, a_s shows periodicity during flapping. This means that the main jitters of video are caused by periodic flapping, which confirms the analysis in Section III-A.

As mentioned in the last subsection, there is a maximum value during a single flapping or jitter cycle. Therefore, the jitter frequency can be estimated by peak detection and periodic segmentation based on a_s . Assuming that a_s has w peaks after detection (peak value threshold is 5), their frame

indexes are $K = \{K_1, K_2, \dots, K_w\}$. The frames between two adjacent peaks are considered as belonging to the same jitter period, having approximately the same jitter frequency. For the t th frame, if there are K_j and K_{j+1} ($j \in [1, w]$) satisfying $t \in (K_j, K_{j+1})$, the jitter frequency f is calculated by:

$$f(t) = \frac{r}{K_{j+1} - K_j}, t \in (K_j, K_{j+1}) \quad (3)$$

while r is the video frame rate. As the original long video is divided into several clips for stabilization, incomplete jitter periods occur at the beginning and end of these video clips. Their jitter frequency is considered the same as that of the adjacent complete jitter periods to ensure that the stabilization can run completely. As the duration is very short, there is not a significant impact on the stabilized results.

The median of f over the entire video is taken as the reference frequency f_B . When part of the frequency is missing, such as during gliding, it is set to f_B virtually to keep the proposed method running, as shown in Fig. 7(a). The slight jitters caused by wind during gliding rather than stopping stabilization can also be removed, which is verified by the video *Flapping to Gliding* and *Gliding to Flapping* in the appendix.

As shown in Fig. 7(a), the estimated frequency occurs in a certain range. It is related to the flight characteristics of the adopted FWR, which needs to keep flapping at some frequency to maintain flight (2~6 Hz in our work, as mentioned in Fig. 2). This frequency range is determined by the mechanical design of the FWR, such as the wings, brushless motor and reduction mechanisms used. Generally, the minimum frequency should not be too low to avoid the flapping being stuck by the friction force. When the FWR stops flapping during gliding and landing, the estimated frequency is set to the reference frequency f_B to keep the stabilization running. On the other hand, the maximum frequency is also limited by the maximum rotation speed of the brushless motor.

To compare the estimated result with the actual flapping frequency, as shown in Fig. 2, a Hall sensor is installed on the reduction mechanism of the FWR, which is triggered during flapping, and the time intervals are recorded. Fig. 7(b) shows the estimated and actual frequencies during climbing. With the time delay eliminated, both have the same trend with an average error of 0.12. They are close enough to verify the assumption that the video jitters are mainly caused by wing flapping.

Although the other sensors on the FWR can also estimate the jitter frequency, such as the Hall sensor, GPS and IMU device, the significant time delay between the camera and these sensors cannot be ignored. Therefore, we still use the proposed method for jitter frequency based on video data only, which still works when other sensors fail.

IV. TRAJECTORY ADJUSTMENT & SMOOTHING

The FWR adjusts the flapping frequency in different flight states, such as climbing, turning and gliding. The jitter frequency and amplitude of the video and its feature trajectories also change, making them challenging to properly smooth with fixed parameters. Therefore, these feature trajectories

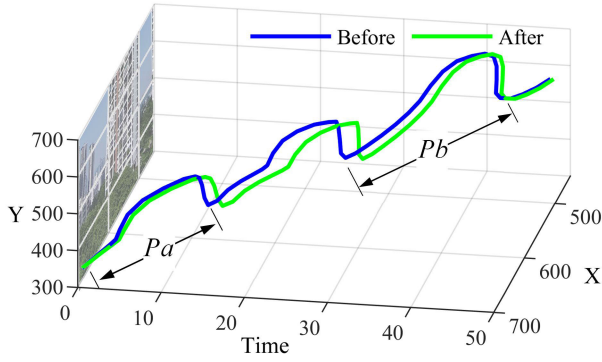


Fig. 8. Comparison of the feature trajectory before and after adjustment.

are adjusted adaptively for jitter frequency consistency by changing their time intervals, and then they are smoothed individually by the sampling-interpolation-averaging strategy.

A. Feature Trajectory Adjustment

The original time interval of each point in a feature trajectory is $1/r$ (r is the video frame rate). Based on the jitter frequency f and reference frequency f_B , the time interval $v(t)$ between the t th and $t + 1$ th frame is adjusted by:

$$v(t) = \frac{f(t)}{f_B} \cdot \frac{1}{r} \quad (4)$$

if $f(t)$ is greater than f_B , $v(t)$ increases. Conversely, when $f(t)$ is smaller than f_B , $v(t)$ decreases. The time interval $v(t)$ is no longer fixed. As shown in Fig. 8, the time of trajectory segment Pa with a short jitter period (high frequency) is extended, while that of segment Pb with a long jitter period (low frequency) is compressed. The jitter frequency of the whole trajectory tends to be consistent with f_B after time interval adjustment.

The original feature trajectory is denoted as P , and its beginning and end frame indexes in the video are tb and te , respectively. The time intervals of P after adjustment are $\{v(tb), v(tb+1) \dots v(te-1)\}$. Then, the feature points in P are linearly sampled for smoothing with a fixed time interval $1/r$. The trajectory after adjustment and sampling is denoted as Q . The above process is called adjustment. This destroys the original time data, and the length of Q may be different from that of P .

The trajectory S is the smoothed result of Q , which must be used to recover the time data P . First, S is sampled according to the adjusted time intervals $\{v(tb), v(tb+1) \dots v(te-1)\}$. Then, the time interval between the sampling points is set to $1/r$, and the trajectory Ps is generated, which has a point-to-point correspondence with P . This process is called the inverse adjustment.

In summary, with adjustment and inverse adjustment, the trajectories at varied jitter frequencies can be converted into states of constant jitter frequency and then be smoothed. Finally, the time intervals of the smoothed results can be recovered, as shown in Fig. 9.

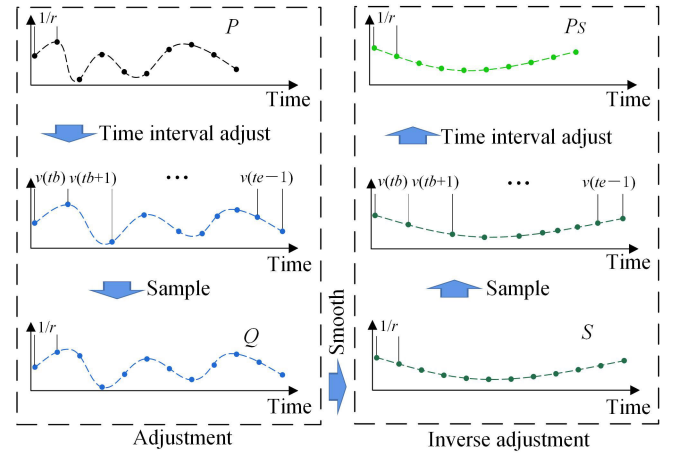


Fig. 9. The process of trajectory adjustment and inverse adjustment.

B. Feature Trajectory Smoothing

The flapping flight of the FWR can be considered the combination of stable gliding and hovering flight with periodic wing flapping. Hence, the feature trajectory after adjustment Q can also be regarded as the combination of a smooth trajectory S and a jitter trajectory with a fixed frequency J :

$$Q(i) = S(i) + A(k) \cdot J(i) \quad (5)$$

where i is the point index of the trajectory. $A(k)$ is the amplitude of J , which varies according to jitter period index k . The jitter frequency of J is f_B . i can also be expressed by:

$$i = n_B \cdot k + h \quad (6)$$

where n_B is the reference jitter period $n_B = r/f_B$, as mentioned in Section III. h is the phase index, $h \in [0, n_B)$. Therefore, $J(i) = J(n_B \cdot k + h) = J(h)$. In the same jitter period, $A(k) \cdot J(h)$ approximately satisfies:

$$\sum_{h=0}^{n_B-1} A(k) \cdot J(h) = 0 \quad (7)$$

Assuming that h is fixed, Q and $S + AJ$ intersect when $k = 0, 1, 2 \dots$ according to (5) and (6). The corresponding points in Q are sampled as $\{Q(h), Q(n_B + h), Q(2n_B + h) \dots\}$. The gaps between these reference points are filled by linear interpolation. Then, the interpolation trajectory E^h is generated with the same trajectory length L as Q , which is related to h . Since E^h is the resampling result of Q , it can be approximately considered that the error between $E^h(i)$ and $S(i)$ is $A(k) \cdot J(h)$ according to (5):

$$S(i) \approx E^h(i) - A(k) \cdot J(h) \quad (8)$$

We obtain all the E^h corresponding to $h \in [0, n_B)$ and substitute them into (7) and (8) for the average:

$$S(i) = \frac{1}{n_B} \sum_{h=0}^{n_B-1} (E^h(i) - A(k) \cdot J(h)) = \frac{1}{n_B} \sum_{h=0}^{n_B-1} E^h(i) \quad (9)$$

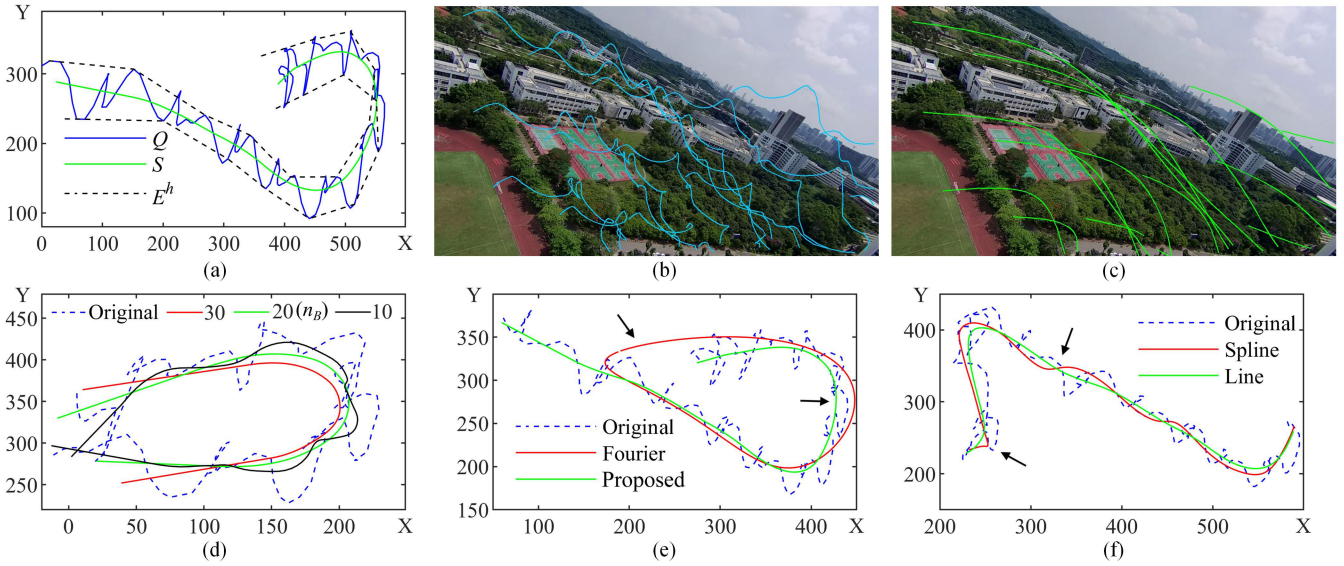


Fig. 10. (a) The smoothing result of a real feature trajectory, only two interpolation trajectory E^h are shown for simplicity. The comparison between real feature trajectories before (b) and after (c) smoothing. (d) Smoothing results with different sampling intervals. (e) The comparison between Fourier transform smoothing and the proposed method. (f) Comparison between linear interpolation and cubic spline interpolation.

S can be calculated from the average of all E^h by (9), which is not affected by $A(k)$ and $J(h)$. The above process is called the smooth strategy of sampling-interpolation-averaging. Fig. 10(a) shows the smoothing process of a real feature trajectory. All the jitters of the original trajectories are smoothed, including the major and minor jitters. The smoothed results are shown in Fig. 10(b) and (c).

The rapid scene change of the video caused by the FWR turning results in many short trajectories. If n_B is used as the fixed sampling interval, it is difficult to sample and smooth the short trajectories with length $L \leq 2n_B$. Removing them may also lead to a lack of data and stabilization failure. With the adaptive sampling interval n_S , most of them could be retained and smoothed. First, short trajectories with length $L \leq n_B$ are considered noise and removed. Then, for trajectories with the length $L > 2n_B$, smoothing performs best with $n_S = n_B$. Finally, for short trajectories with the length $L \in (n_B, 2n_B]$, n_S is:

$$n_S = \left\lceil \frac{L - 1}{2} \right\rceil \quad (10)$$

where $\lceil \cdot \rceil$ is the rounding function. To smooth these short trajectories, n_S in (10) is not only adjusted adaptively according to the trajectory length but also as close to n_B as possible for better performance. The smoothing results with different sampling intervals are shown in Fig. 10(d). If the sampling interval deviates from n_B , the results lack smoothing or are oversmoothed, while n_B is determined by the reference frequency without any artificial adjustment.

As described in the last subsection, the smoothing result S should be adjusted inversely for PS, which has a point-to-point correspondence with P. Finally, with the guidance of P and PS, the CPWs in [21] are used to generate stabilized video.

Finally, the proposed method is summarized as algorithm 1.

C. Comparison & Discussion

There are several comparisons and discussions about different smoothing and interpolation methods.

1) *Smoothing Method*: Fourier transform smoothing with the minimum jitter frequency as the cutoff frequency is compared. As shown in Fig. 10(e), Fourier transform smoothing results in abnormal warping and noticeable deviations (as indicated by the arrows), which may induce unexpected distortion in the output video, while the proposed method does not deviate from the original trajectory in the middle or at the end.

2) *Interpolation Method*: The cubic spline interpolation is compared, which can generate smoother interpolation trajectories than the linear interpolation. However, it also induces unnecessary bending because too many points need to be interpolated between each reference point (approximately 10 to 30 points in this work), such as in the significant jitters and distortion shown in Fig. 10(f). The smoothed result of cubic spline interpolation has obvious jitters and warping (as indicated by the arrows). By linear interpolation, the individual interpolation trajectories may not be sufficiently smooth, but their average is. In addition, cubic spline interpolation requires at least three reference points, while linear interpolation requires only two, which means that linear interpolation can handle shorter trajectories.

V. EVALUATION & COMPARISON OF STABILIZATION

The monitoring videos are captured by an airborne vision system on the FWR during an outdoor flight test in the airspace 20 m to 100 m above the ground. Several video samples are extracted from the original videos, which are stabilized by the proposed method and other existing academic and commercial methods for comparison. Then, objective and subjective evaluations are applied to these stabilization results.

Algorithm 1 Video Stabilization

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 $r \leftarrow$  the video frame rate
 $P \leftarrow$  the original feature trajectory generated by the KLT tracker
 $Q \leftarrow$  the trajectory after time interval adjustment based on  $P$ 
 $S \leftarrow$  the smoothed trajectory based on  $Q$ 
 $P_S \leftarrow$  the smoothed trajectory after inverse adjustment based on  $S$ 
for each frame  $t$  do
     $a(t) \leftarrow$  calculated by (2)
end for
 $a_S \leftarrow$  the smoothing result of  $a$ 
 $K \leftarrow$  peak detection based on  $a_S$ 
for each frame  $t$  do
     $f(t) \leftarrow$  calculated by (3)
end for
 $f_B \leftarrow$  the median of  $f$ 
 $n_B \leftarrow r / f_B$ 
for each frame  $t$  do
     $v(t) \leftarrow$  calculated by (4)
end for
for each trajectory  $P$  do
     $tb, te \leftarrow$  the starting and ending frame index of  $P$ 
     $Q \leftarrow$  adjusted and sampled based on  $P$  and  $v(tb) \dots v(te-1)$ 
     $L \leftarrow$  the length of  $Q$ 
     $n_S \leftarrow n_B$ 
    if  $L \in (n_B, 2n_B]$  then  $n_S \leftarrow$  calculated by (10)
    for each phase  $h \in [0, n_S]$  do
         $E^h \leftarrow$  interpolated based on  $Q(h), Q(n_S + h), Q(2n_S + h) \dots$ 
    end for
     $S \leftarrow$  calculated by (9) and all  $E^h$ 
     $P_S \leftarrow$  resampled based on  $S$  and  $v(tb) \dots v(te-1)$  and adjusted inversely.
end for
Output the stabilized video by CWP with all  $P$  and  $P_S$ 

```

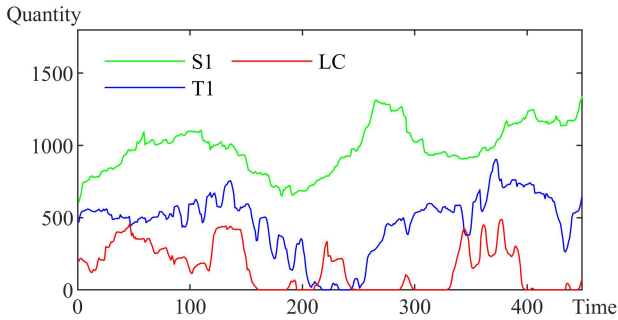


Fig. 11. The number of long trajectories at each time in typical video samples.

A. Video Samples

According to the flight path and state of the FWR, 15 video samples are classified as straight, turning or unsteady flight. In straight flight, the FWR flies along a roughly straight line, and there are 6 video samples marked as S1 ~ S6. In turning flight, there are 6 video samples marked as T1 ~ T6. The video samples of unsteady flight refer to different flight states of the FWR, including the transformation between flapping and gliding (marked as FtG and GtF) and the lost control state with a large pitch angle (marked as LC). The captures of these samples are shown in Fig. 12. It is recommended to watch these video samples directly by following the links in the appendix.

The original feature trajectories of all video samples are generated by the KLT tracker. To evaluate the speed of

scene change in typical video samples, the number of long trajectories greater than 40 in length (referred from [18]) is counted per frame, as shown in Fig. 11. It is found that there are many long trajectories when the video scene changes slowly, such as in video S1. During quick turning of the FWR, there is a lack of long trajectories at some time in video T1. Video LC is an extreme case because the number of long trajectories is zero many times.

Notably, it is difficult to capture videos with rich scenes at close range, such as with handheld cameras, because of safety. On the other hand, as the scenes in aerial videos are mainly buildings and vegetation, moving objects or large areas of occlusion are not accounted for.

B. Versus Methods

Several traditional DVS methods are implemented as versus methods, including [18], [23], and [24], which are named *Subspace*, *Augment* and *Bundle*. These methods use the same original feature trajectories as the proposed method. Commercial methods are also adopted, including Adobe After Effects CC, Final Cut Pro X and VirtualDub Deshaker, which are named *AE*, *FCPX* and *Deshaker*, respectively. From the DVS works based on deep learning in recent years, three representative methods are selected, including *StabNet* [30], *DIFRINT* [33], and *DUT* [34]. The parameters of these methods are set to the recommended or default values. Although these methods are not specifically developed for airborne vision of FWR, they still represent advanced DVS.

C. Evaluation & Results

Quantitative and user evaluation techniques are applied to the stabilization results of all methods. The quantitative evaluation includes three ratios: cropping, distortion and stability, which have been adopted by other studies such as [23], [24], and [37]. User evaluation is also a reliable subjective method to evaluate the quality of stabilized videos in [38].

1) Quantitative Evaluation:

a) *Cropping*: Since irregular blank areas appear at the video's border after stabilization, they should be cropped by the maximum rectangular border for visual comfort. The cropping ratio is the ratio of the pixel area between the original and stabilized video. A higher cropping ratio means that more pixel area is retained after stabilization.

b) *Distortion*: The global homography transformation between the original and stabilized frame is calculated, and the ratio of the two maximum eigenvalues of its affine part is taken as the distortion ratio of this frame. For all frames, the minimum distortion ratio is the final output. The higher the distortion ratio is, the less distortion the stabilization result has.

c) *Stability*: The global jitter acceleration sums of the original and stabilized video are calculated separately, as mentioned in Section III. The stability ratio is the ratio of the reduction between them. A higher stability ratio means fewer jitters in the stabilization result.

These ratios are close to 1 when the stabilization result is satisfactory. The quantitative evaluation results are shown

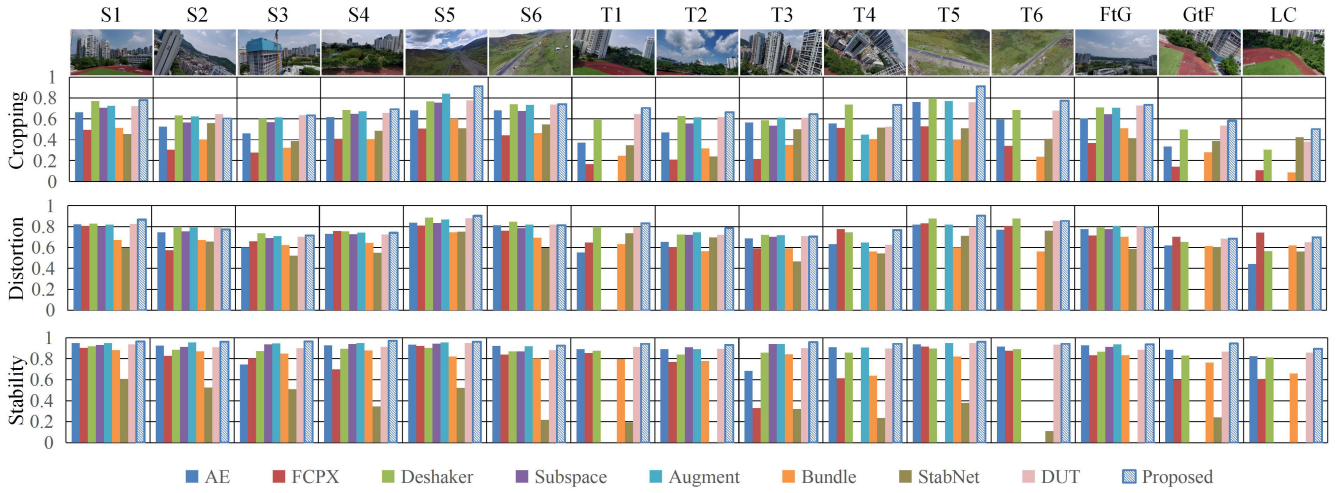


Fig. 12. Objective evaluation results.

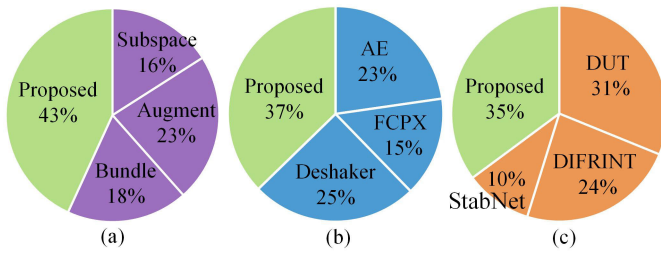


Fig. 13. Subjective evaluation results of traditional (a), commercial (b), and deep learning (c) group.

in Fig. 12. If a method fails to stabilize a video sample, its evaluation result is missing. Furthermore, the mean values of the cropping, distortion and stability ratios are also shown in Table II to illustrate the overall advantages of the proposed method.

2) *User Evaluation*: Forty participants were invited to the user evaluation. The stabilization methods are divided into three groups: traditional, commercial and deep learning, which are compared separately to avoid participant fatigue. In each group, participants were shown several stabilization results, randomly and blindly named A, B, C and D. They were asked to pay attention to cropping, distortion and stability and rank the results from excellent to poor according to subjective feeling, e.g., $B > D > A > C$. The total score for each video sample was 10. The best one to the worst received scores of 4 to 1, respectively. The method received a score of 0 if it failed. Finally, the total score ratio in all video samples of each method was calculated. A method with a higher score ratio was more popular among the participants. The subjective evaluation results are shown in Fig. 13.

D. Analysis & Discussion

As shown in Table III, the proposed method is more robust than the *subspace* and *augmentation* methods, which failed four times and three times, respectively, especially in cases that lacked long feature trajectories, such as turning and

unsteady flight cases. With the advantage of smoothing short trajectories, the proposed method retained enough data for stabilization, which succeeded in all samples.

In addition to the *subspace* and *augmentation* methods, the proposed method was based on the smoothing of the feature trajectories. In the *subspace* method, all the trajectories are decomposed into two low-rank matrices by a sliding window, these matrices were smoothed, and then the smoothed trajectories were reconstructed. The *augmentation* method minimizes the first to third order derivatives of the trajectories for smoothing based on a linear programming framework. These two methods both rely on sliding sampling windows, and complete trajectories in the window should be sufficient; otherwise, they may fail. The proposed method has the advantage that all trajectories are smoothed individually instead of globally, which reduced the failure caused by the lack of long trajectories. Second, the proposed method has more potential for efficiency without dependence on matrix decomposition or least squares optimization.

In cropping, the proposed method retains more pixel area than the *FCPX* method while having less distortion than the *Bundle* method. Instead of the mesh trajectories in the *Bundle* method, many feature trajectories are used to stabilize videos in the proposed method, which can better describe the scene motion. Therefore, it induces fewer blank areas and distortion in the stabilization results, especially in video samples of turning and unsteady flight, as shown in Fig. 14.

The *AE* and *Deshaker* methods are classic commercial methods that perform comparably to the proposed method in most cases. However, in several extreme cases, such as videos GtF and LC, the proposed method has more advantages in cropping and stability. For the best performance in various videos, the parameters have to be adjusted with other methods, while the proposed method can stabilize videos adaptively without any artificial adjustment due to the jitter frequency estimation.

On the other hand, in the subjective evaluation, the score ratio of the proposed method is the highest in all groups. In the

TABLE II
THE AVERAGE OF CROPPING, DISTORTION & STABILITY

	S1	S2	S3	S4	S5	S6	T1	T2	T3	T4	T5	T6	FtG	GtF	LC
AE	0.811	0.731	0.602	0.758	0.816	0.804	0.604	0.671	0.646	0.699	0.837	0.759	0.770	0.614	0.421
FCPX	0.735	0.568	0.581	0.622	0.746	0.681	0.556	0.528	0.379	0.635	0.758	0.675	0.638	0.481	0.485
Deshaker	0.838	0.772	0.737	0.777	0.852	0.818	0.758	0.730	0.722	0.780	0.859	0.818	0.789	0.659	0.560
Subspace	0.811	0.743	0.731	0.770	0.843	0.776	/	0.729	0.726	/	/	/	0.778	/	/
Augment	0.832	0.791	0.755	0.788	0.888	0.823	/	0.751	0.756	0.667	0.845	/	0.817	/	/
Bundle	0.689	0.648	0.598	0.644	0.722	0.652	0.559	0.555	0.595	0.535	0.609	0.194	0.681	0.552	0.455
StabNet	0.554	0.580	0.472	0.458	0.594	0.453	0.425	0.220	0.430	0.431	0.534	0.427	0.354	0.410	0.332
DUT	0.826	0.782	0.746	0.763	0.868	0.812	0.782	0.744	0.739	0.683	0.834	0.820	0.804	0.696	0.628
Proposed	0.870	0.780	0.772	0.801	0.926	0.825	0.826	0.792	0.769	0.814	0.926	0.856	0.821	0.737	0.698

TABLE III
COMPARISON OF SUCCESS RATE

Method	Subspace	Augment	Proposed
Success rate	60% (9/15)	73.33% (11/15)	100% (15/15)



Fig. 14. Compared with the proposed method, method FCPX has obvious cropping and amplifying, while method Bundle has abnormal distortion.

traditional group, its score is significantly higher than others due to the failures of other methods. The feedback from the participants also suggests that the other methods were ranked at the bottom, while their results show obvious shortcomings in cropping, distortion or stability, e.g., the method *FCPX*, which preforms poorly in cropping. There is no significant weakness in the proposed method, which has a good overall performance among all the methods.

The *Bundle* method stabilized video by using a bundle of multiple camera paths and space-time path optimization, which has difficulty achieving a trade-off between stability and cropping ratio. Because of space-time bilateral smoothing, the main motion of the video may be oversmoothed, resulting in a large blank area. Instead of smoothing the whole motion, the proposed method smooths the jitter while preserving the main motion.

Furthermore, the proposed method is compared with the *Bundle* method in video motion paths and cropping ratio. According to [24], the motion paths of the original and stabilized video are generated, and then the cropping ratios per frame are calculated. Although the motion path of the

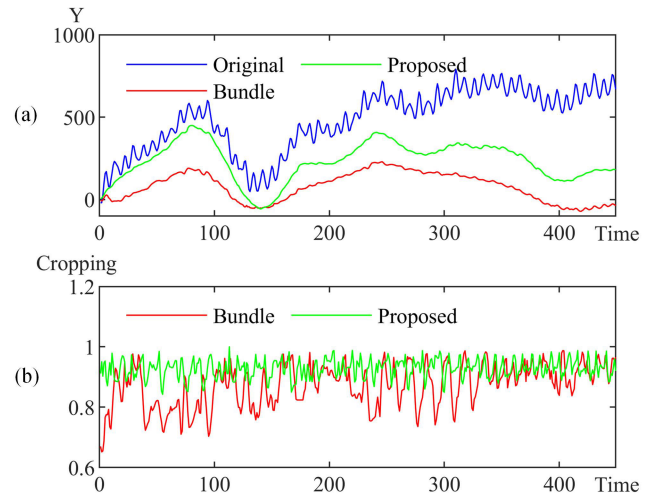


Fig. 15. The comparison between Bundle and proposed in video motion path (a) and cropping rate (b).

proposed method fluctuates more than that of the *Bundle* method, its cropping ratio is much higher as shown in Fig. 15, which means that the proposed method preserves the main motion of the video rather than weakening it.

The *StabNet*, *DIFRINT*, and *DUT* methods are all based on deep learning. The convolutional neural network model (*StabNet*) can stabilize video in real time, but many labeled datasets are essential for training. The trajectory could be smoothed by a dynamic smoothing kernel (*DUT*) or the interframe interpolation model (*DIFRINT*) with unsupervised training, but the time and space complexity of these methods are generally poor and depend on large computational power. In contrast, the proposed method is much simpler and more efficient, benefiting from trajectory smoothing by interpolation and averaging.

Although *DIFRINT* has the advantage of full frames without cropping, it produces obvious ghost artifacts in all video samples, while the proposed method did not, as shown in Fig. 16(a). Ghost artifacts may increase the risk of error in measurement and detection. Since there is no quantitative evaluation of ghost artifacts in academia, *DIFRINT* does not participate in the objective evaluation for fairness. From the subjective evaluation, the score rate of the proposed method (35%) is higher than that of *DIFRINT* (24%) due to ghost artifacts.

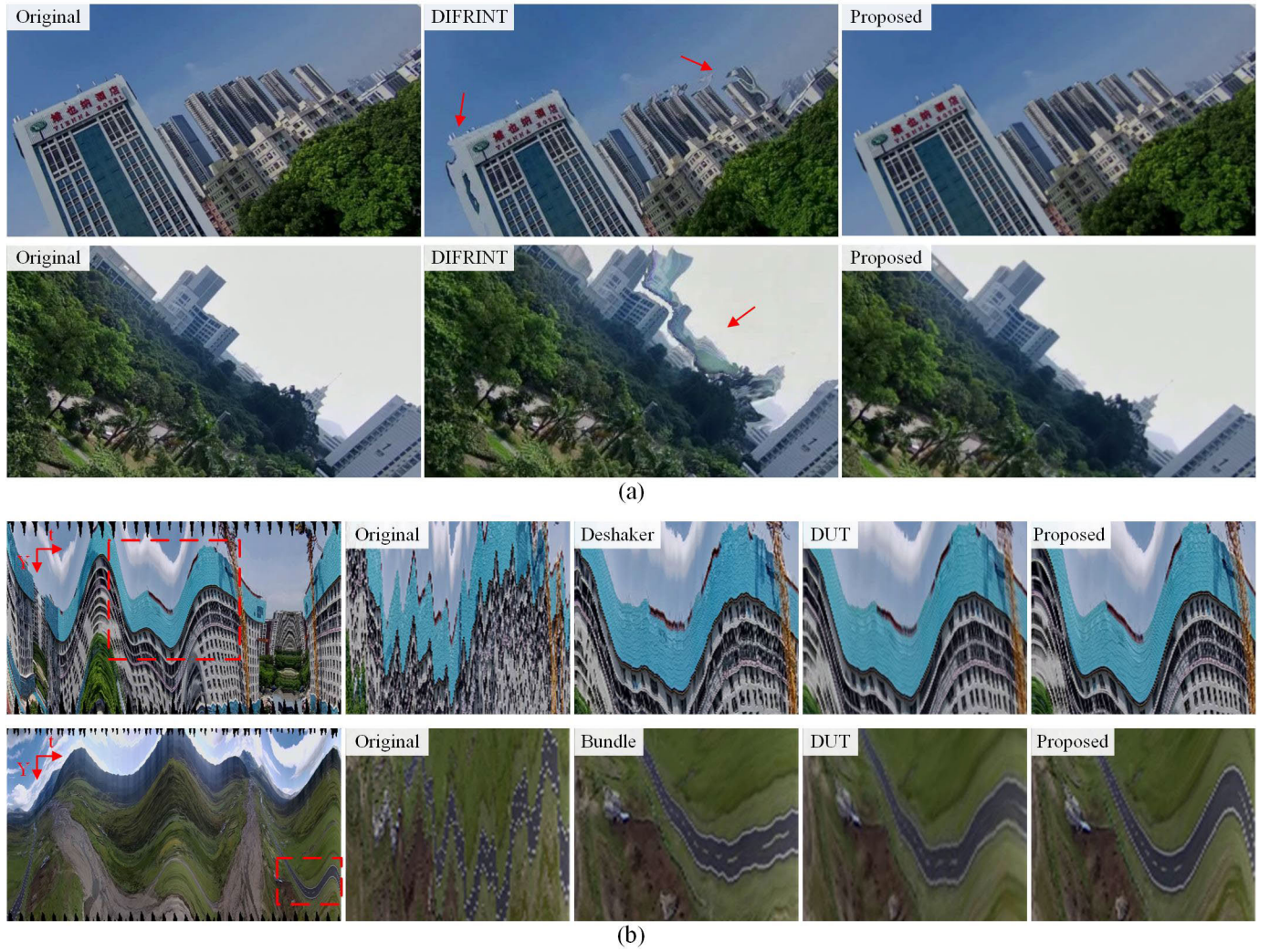


Fig. 16. (a) The ghost artifacts produced by DIFRINT. (b) The comparison of EPI.

In the quantitative evaluation, the proposed method has obvious advantages over *StabNet* and slight advantages over *DUT*. In subjective evaluation, the score rate of the proposed method (35%) is also higher than that of *StabNet* (10%) and *DUT* (31%), which agrees with the results of the quantitative evaluation. Furthermore, the epipolar plane images (EPIs) of video samples are generated. The streamlines in the EPI (such as the building and road) reflect the movement of pixels in the Y direction over time. The smoother the streamlines are, the fewer jitters the video has. Fig. 16(b) shows that the result of the proposed method has fewer residual jitters compared to *DUT*, *Deshaker* and *Bundle*. Notably, the proposed method does not rely on a large amount of training data or GPU, which is easier to implement onboard and in real-time in FWRs.

E. Robustness Test

The proposed method is also tested with low-quality samples, including night-scene, noisy and blurry samples, as shown in Fig. 17. Although insufficient illumination, noise and blur may reduce the success rate of the KLT tracker and make it difficult to form long trajectories, the proposed method stabilizes these samples successfully due to its ability



Fig. 17. The low-quality samples and a extreme case of failure.

to process short trajectories. We also show a case of failure, which is due to an extreme lack of feature points in the low-textured and insufficient illumination area of night sky with noise and blur.

In these states with insufficient illumination, noise or blur, long trajectories are difficult to generate because of the poor success rate of the KLT tracker. Although the proposed method can smooth some short trajectories, it still fails in cases in

which most of the trajectories are very short (such as less than 20 in our work), as the failure scenes with low texture and lack of lighting shown in Fig. 17. The proposed method still relies on the robust feature point tracker, as do other methods based on feature trajectories.

On the other hand, the proposed method may also fail in scenes of dynamic objects with large sizes, such as walking people at close range. The jitter frequency estimation is inaccurate due to the unexpected object motion. However, we focus on video shooting by FWR in the air, and most of the objects in the video are static, such as vegetation and buildings. Dynamic objects such as moving people or cars are much smaller. As a result, these objects at close range have not yet been considered or tested due to safety concerns.

F. Running Time & Long Video

All methods (except method *FCPX*) are implemented on a computer with a 1.7 GHz Intel Core i5 CPU and 4 GB RAM. The proposed method is developed in C++ based on OpenCV. Its time consumption for feature trajectory generation, trajectory smoothing, and frame warping is 400 ms, 20 ms, and 230 ms per frame without additional optimization for computational speed, respectively. The feature point number is set to 1000 per frame, and the grid number of frame warping is set to 5×5 . In trajectory smoothing, the proposed method can be implemented easily by multithreading. For comparison, the time consumption of trajectory smoothing in the subspace method is 5 ms per frame, which is the lowest among all methods. Feature trajectory generation and frame warping are the main bottlenecks of computational performance in the proposed method. If the video resolution can be reduced, the running time will improve significantly, such as 24.6 FPS at 640×360 video resolution.

On the other hand, stabilizing the entire monitoring video directly would require huge computing resources. Usually, a long video would be divided into several subvideos to be stabilized separately and then combined. It is recommended that the overlap length between each subvideo be at least $2n_B$, as mentioned in Section IV. Then, the beginning and end n_B frames of the stabilized subvideo should be removed before combining the subvideos. Two 30-second videos, which are stabilized by the proposed method and the above process, are shown with links in the appendix.

VI. CONCLUSION

With the help of jitter frequency estimation and adaptive trajectory adjustment, the proposed method performs video stabilization in different flight states of FWRs without pan-tilt camera assistance. Compared with the existing methods, it shows better robustness, cropping and stability, which enhance the application potential of FWR with airborne vision. However, in scenes of low-textured or insufficient illumination, such as night skies, grassland, sea surfaces and snowfields, it may fail due to the extreme lack of features. Unlike for the gimbal camera, the loss of field of view is unavoidable with the proposed method due to cropping. Due to the strict requirements on the load and power consumption

TABLE IV
THE WEBSITE LINKS OF THE VIDEO CLIPS MENTIONED IN THE PAPER

Video	Link
Straight	S1 https://youtu.be/zFbE81YdJNl
	S2 https://youtu.be/oCrgsOLVOrs
	S3 https://youtu.be/k3HGuhV8n1s
	S4 https://youtu.be/6BpYVRhbG_0
	S5 https://youtu.be/nzjM1TL_fGA
	S6 https://youtu.be/RANx1kSqhMg
Turning	T1 https://youtu.be/0k5tDnUonr8
	T2 https://youtu.be/zejUUtZ2Tsg
	T3 https://youtu.be/Tyk3FSMOAsI
	T4 https://youtu.be/EGYc1vg03PA
	T5 https://youtu.be/yfjjlvpxr54
	T6 https://youtu.be/plV1GODv5jA
Flapping to Gliding	https://youtu.be/6alygDSNy6w
Gliding to Flapping	https://youtu.be/_i13N7YG9q8
Lost Control	https://youtu.be/Vq0QuZPfOiQ
Long Video1	https://youtu.be/jBMPrltW2Ik
Long Video2	https://youtu.be/MC2CXqmS9_Y
Low--quality	https://youtu.be/4oWiaLWT02M
Gimbal Camera	https://youtu.be/8Y1McpOaSEI

of FWRs, onboard and real-time processing have not yet been implemented.

In the future, a control model of FWR will be established, and the jitters of airborne vision will be reduced by body stability control based on visual feedback. The feature trajectory of periodic jitters will be predicted with the help of historical jitters to improve the accuracy of jitter frequency estimation. The proposed method is also suitable for high-speed parallel computing on embedded chips (such as FPGA). After the improvement of the load capacity of the FWR and carrying the high-performance embedded chip, video stabilization with the proposed method can be completed onboard online. The stabilized results can be transmitted to the ground station for further visual processing, reducing the processing burden on the ground and improving real-time performance.

APPENDIX

See Table IV.

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